OPENNESS OF THE FLAT LOCUS IN THE FINITE TYPE CASE, VIA THE FINITE PRESENTATION THEOREM

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The finite presentation version of openness of flatness. Same chapter flat.tex, label theorem-finite-type-flat, invoked at proof line 4250. Applied to the ambient affine space over the base and the pushforward of the module, it gives that the flat locus there is open and that the restriction of the pushforward is of finite presentation.
- Relative finite presentation and its behaviour under a finite morphism. More on Morphisms, labels more-morphisms-definition-relatively-finitely-presented-sheaf (statement, proof line 4243), more-morphisms-lemma-relative-finite-presentation and more-morphisms-lemma-finite-morphism-relative-finite-presentation (proof lines 4266-4267). Supplies the notion of being locally finitely presented relative to the base and shows that it is inherited from the pushforward along the closed immersion.

Source reference. Original definition 05H1: sources/stacks-chow/tags.txt, source lines 7080–7080. Navigation pointer for \ref{more-morphisms-definition-relatively-finitely-presented-sheaf} used inside the counted span (line 4244); the argument that consumes it is retained in the span.

Source reference. Original lemma 09T9: sources/stacks-chow/tags.txt, source lines 12382–12382. Navigation pointer for \ref{more-morphisms-lemma-finite-morphism-relative-finite-presentation} used inside the counted span (line 4267); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 09T8: sources/stacks-chow/tags.txt, source lines 12381–12381. Navigation pointer for \ref{more-morphisms-lemma-relative-finite-presentation} used inside the counted span (line 4266); the argument that consumes it is retained in the span.

Source reference. Original theorem 05IK: sources/stacks-flat/pinned-original-source.tex, source lines 4097–4097. Navigation pointer for \ref{theorem-finite-type-flat} used inside the counted span (line 4230); the argument that consumes it is retained in the span.

Lemma 0.1 (Finite type version of Theorem 05IK)**.** *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Assume*

(1) *$X \to S$ is locally of finite type,*
(2) *$\mathcal{F}$ is an $\mathcal{O}_X$-module of finite type, and*
(3) *the set of weakly associated points of S is locally finite in S.*

Then $U = \{x \in X \mid \mathcal{F}$ flat at x over $S\}$ is open in X and $\mathcal{F}|_U$ is flat over S and locally finitely presented relative to S (see More on Morphisms, Definition 05H1).

Proof. The question is local on X and S. Thus we may assume X and S are affine. Then we may choose a closed immersion $i : X \to \mathbf{A}^n_S$. We apply Theorem 05IK to $X' = \mathbf{A}^n_S \to S$ and the quasi-coherent module $\mathcal{F}' = i_*\mathcal{F}$ of finite type and we find that

$$U' = \{x' \in X' \mid \mathcal{F}' \text{ flat at } x' \text{ over } S\}$$

is open in X' and that $\mathcal{F}'|_{U'}$ is of finite presentation. Since $\mathcal{F}'$ restricts to zero on $X' \setminus i(X)$ and since $\mathcal{F}'_{i(x)} \cong \mathcal{F}_x$ for all $x \in X$ we see that

$$U' = i(U) \amalg (X' \setminus i(X))$$

Hence $U = i^{-1}(U')$ is open. Moreover, it is clear that $\mathcal{F}'|_{U'} = (i|_U)_*(\mathcal{F}|_U)$. Hence we conclude that $\mathcal{F}|_U$ is finitely presented relative to S by More on Morphisms, Lemmas 09T8 and 09T9. □